UNIVERSITY of HOUSTON | ECE

ECE 3337: Test #2

Spring 2023

Last name of student	Student ID number
Solutions	
First name of student	**Email address**

Do This First:

1. Make sure that you have all the pages of the exam.
2. Fill in the above information.
3. Turn off and put away <u>all</u> electronic gadgets (laptop, tablet, smartphone, calculator, smart watch, …)

Exam Rules:

1. You are allowed to bring one 8.5” x 11” handwritten crib sheet (double sided).
2. Indicate your answers in the spaces provided. Use the back of the page to carry out calculations.

Q1	Q2	Q3	Q4	Q5	Q6	Total
20	20	25	15	10	10	100
Laplace Transform	Inverse Laplace Transform	Circuit Analysis	Response Partitions	Op Amp Circuits	Block Diagrams	

1. (20 points) Laplace Transforms

a. (6 points) A signal is given by $x(t) = r(t) - 2r(t-2) + 2(t-5)u(t-5) - (t-6)u(t-6)$. Find $\mathbf{X}(\mathbf{s})$, the Laplace transform of $x(t)$.

1 point for trying

$$x(t) = r(t) - 2r(t-2) + 2r(t-5) - r(t-6)$$

$$r(t) = t\,u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s^2} \quad - 1$$

$$r(t-2) \xleftrightarrow{\mathcal{L}} \frac{1}{s^2} e^{-2s} \quad - 1$$

$$r(t-5) \xleftrightarrow{\mathcal{L}} \frac{1}{s^2} e^{-5s} \quad - 1$$

$$r(t-6) \xleftrightarrow{\mathcal{L}} \frac{1}{s^2} e^{-6s} \quad - 1$$

$$x(t) \xleftrightarrow{\mathcal{L}} X(s) = \frac{1}{s^2} - \frac{2}{s^2} e^{-2s} + \frac{2}{s^2} e^{-5s} - \frac{1}{s^2} e^{-6s}$$

Full correct answer 1

(b) (4 points) Given a signal $x_2(t)$, find the Laplace transform of $y(t) = x_2(3(t-1))$ in terms of $\mathbf{X_2}(\mathbf{s})$

1 point for trying

$$x_2(t) \xleftrightarrow{\mathcal{L}} X_2(s)$$

$$x_2(3t) \xleftrightarrow{\mathcal{L}} \frac{1}{3} X_2\left(\frac{s}{3}\right) \quad 1 \quad 1$$

Scaling property

$$x_2(3(t-1)) \xleftrightarrow{\mathcal{L}} \frac{1}{3} X_2\left(\frac{s}{3}\right) e^{-s}$$

1

Time shifting

c. (10 points) Find the Laplace transform of $z(t) = t\, e^{-9t} \sin(3t)\, u(t)$, and find the values of all its poles and zeros.

2 points for trying

$$\sin 3t \; u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{3}{s^2 + 3^2} = \frac{3}{s^2+9} \quad — \; 1$$

$$e^{-9t} \sin 3t \; u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{3}{(s+9)^2 + 9} \quad — \; 1$$

$$t \; e^{-9t} \sin 3t \; u(t) \overset{\mathcal{L}}{\longleftrightarrow} -\frac{d}{ds}\left[\frac{3}{(s+9)^2+9}\right] \quad — \; 1$$

$$= \frac{3 \cdot 2(s+9)}{\left[(s+9)^2 + 9\right]^2}$$

$$= \frac{6\,(s+9)}{\left[(s+9)^2 + 9\right]^2} \quad — \; 1$$

Zeros : $s = -9$ — 1

Poles $\left[(s+9)^2 + 9\right]^2 = 0$ — 1

Repeated roots $(s+9)^2 = -9 = -3^2$

$(s+9) = \pm j3$

$s = -9 \pm j3$ — 1

Poles : 1 $\begin{cases} -9 + j3, \\ -9 + j3, \\ -9 - j3, \\ -9 - j3 \end{cases}$

$-9 + j3$, $-9 + j3$ } Repeated

$-9 - j3$, $-9 - j3$ } Repeated

2. (20 points) Inverse Laplace Transform

Find the inverse Laplace transforms for the following systems.

a. (10 points) $X(s) = \frac{e^{-s}(s+4)}{(s^2+8s+20)}$

2 points for trying

$$X(s) = e^{-s} \frac{s+4}{s^2+8s+16+4}$$

$$= e^{-s} \frac{(s+4)}{(s+4)^2+2^2} \quad — 2$$

$$\frac{s}{s^2+2^2} \overset{\mathcal{L}}{\longleftrightarrow} \cos 2t \; u(t) \quad — 2$$

$$\frac{s+4}{(s+4)^2+2^2} \overset{\mathcal{L}}{\longleftrightarrow} e^{-4t} \cos 2t \; u(t) \quad — 1$$

$$e^{-s} \frac{(s+4)}{(s+4)^2+2^2} \overset{\mathcal{L}}{\longleftrightarrow} \underbrace{e^{-4(t-1)}}_{1} \; \underbrace{\cos 2(t-1)}_{1} \; \underbrace{u(t-1)}_{1}$$

Full points for the correct answer

b. (10 points) $X(s) = \frac{12}{(s^3+4s^2+3s)}$

2 points for trying

$$X(s) = \frac{12}{s(s^2+4s+3)} \quad — 1$$

$$= \frac{12}{s(s+1)(s+3)} \quad — 1$$

$$= \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+3} \quad — 1$$

$$A = \frac{12}{3} = 4 \quad — 1$$

$$B = \frac{12}{2} = 6 \quad — 1$$

$$C = \frac{1}{6} \times 12 = 2 \quad — 1$$

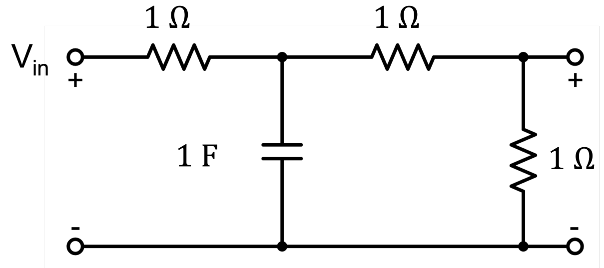
$$X(s) = 4 \cdot \frac{1}{s} + 6 \cdot \frac{1}{s+1} + 2 \cdot \frac{1}{s+3}$$

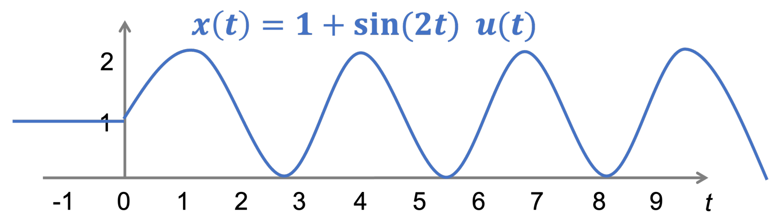
$$x(t) = \underbrace{4\, u(t)}_{1} + \underbrace{6\, e^{-t} u(t) + 2 e^{-3t}}_{1} u(t)$$

Full points for correct answer.

3. (25 points) Circuit Analysis

For the circuit below, the input $V_{in}(t) = x(t)$ is given in the graph. The units for $x(t)$ is Volts [V].

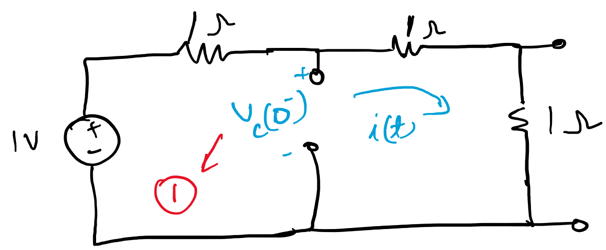




a. (5 points) Find all initial conditions in the time domain for the circuit. Label them on the diagram, making polarities clear.

For $t < 0$, $V_{in}(t) = x(t) = 1\,V$ (1)

1 point for trying

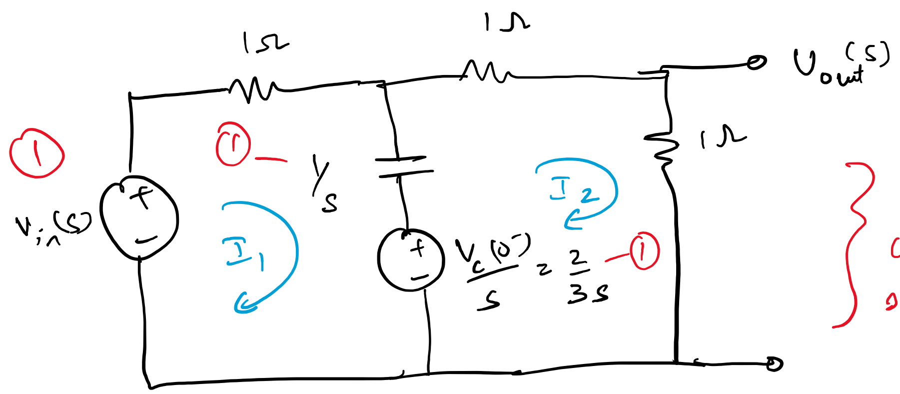


$i(t) = \frac{1V}{(1+1+1)\Omega} = \frac{1}{3}$ A — (1) $\therefore$ $V_c(0^-) = 1 - \frac{1}{3}\cdot 1$

$= \frac{2}{3}$ V — (1)

b. (5 points) Draw the **s**-domain representation of the circuit, including initial conditions.

1 point for trying



circuit schematic (1)

Full points for full correct circuit

3 points for trying

c. (15 points) Find the zero-state response $\mathbf{V_{ZSR}(s)}$. Your answer should be expressed as a ratio of polynomials. There is no need to perform partial fraction expansion.

for ZSR, the initial conditions are zero 1

Applying KVL : $V_{in} - I_1 - (I_1 - I_2)\frac{1}{s} = 0$ — 1

1

$$I_1\left(1+\frac{1}{s}\right) - \frac{I_2}{s} = V_{in} \quad — 1$$

$$I_1(1+s) - I_2 = s V_{in}$$

$$+(I_2 - I_1)\frac{1}{s} + I_2 + I_2 = 0 \quad — 1$$

$$I_1\left(-\frac{1}{s}\right) + \left(2+\frac{1}{s}\right) I_2 = 0 \quad — 1$$

$$-I_1 + (2s+1) I_2 = 0$$

$$\left.\begin{pmatrix} 1+s & -1 \\ -1 & 2s+1 \end{pmatrix}\begin{pmatrix} I_1 \\ I_2 \end{pmatrix} = \begin{pmatrix} s V_{in} \\ 0 \end{pmatrix}\right\} \quad 1$$

$V_{in}(t) = (1+\sin 2t)u(t)$ for $t > 0$

$$V_{in}(s) = \frac{1}{s} + \frac{2}{s^2+4} = \frac{s^2+2s+4}{s(s^2+4)} = \frac{s^2+2s+4}{s(s^2+4)}$$

Determinant $\Delta = \overset{1}{(1+s)(2s+1) - 1} = 2s^2+3s$

$$\Delta_{I_2} = \begin{vmatrix} 1+s & sV_{in} \\ -1 & 0 \end{vmatrix} = \underset{1}{(sV_{in})(1)}$$

$$= sV_{in}$$

$$I_2 = \frac{\Delta_{I_2}}{\Delta} = \frac{s^2+2s+4}{(s^2+4)(2s^2+3s)} \quad — 1$$

$$V_{ZSR} = I_2 R = I_2$$ (1)

$$= \frac{s^2+2s+4}{(s^2+4)(2s^2+3s)}$$ (1)

4. (15 points) System Response Partitions

A linear time-invariant system is described by the following LCCDE

$$\frac{d^2y(t)}{dt^2} + 2\frac{dy(t)}{dt} + y(t) = x(t)$$

The input is $x(t) = e^{-3t}u(t)$, with the initial conditions $y(0-) = 0,\ y'(0-) = 1$.

a. (5 points) Derive a formula for the total output in the Laplace domain. Your answer should be expressed as a ratio of polynomials with a fully factored denominator.

1 point for trying

$$y(t) \xleftrightarrow{\mathcal{L}} Y(s)$$

$$\frac{dy}{dt} \xleftrightarrow{\mathcal{L}} sY(s) - y(0^-) = sY(s) \quad [\because y(0^-) = 0]$$ (1)

$$\frac{d^2y}{dt^2} \xleftrightarrow{\mathcal{L}} s^2Y(s) - s\,y(0^-) - y'(0^-) = s^2Y(s) - 1 \quad [\because y'(0^-) = 1]$$ (1)

The LCCDE transforms to : $s^2Y(s) - 1 + 2sY(s) + Y(s) = X(s)$

$$Y(s)[s^2+2s+1] = X(s) + 1$$

$$Y(s) = \frac{X(s)+1}{s^2+2s+1}$$

$x(t) = e^{-3t}u(t),\ X(s) = \frac{1}{s+3}$ (1) $\therefore$ $Y(s) = \frac{\frac{1}{s+3}+1}{(s+1)^2} = \frac{(s+4)}{(s+1)^2(s+3)}$ (1)

b. (5 points) Calculate the forced response of this system in the Laplace domain.

The modes of the system correspond to the roots of the characteristic equation $(s+1)^2 = 0$ i.e $s = -1, -1$ [modes]

1 point for trying

(1)

The forced response corresponds to the terms other than the modes

i.e. $s+3$ — 1

Let $\dfrac{s+4}{(s+1)^2(s+3)} = \dfrac{A}{s+3} + \dfrac{B_1}{(s+1)} + \dfrac{B_2}{(s+1)^2}$ 1

We just need to find A. $A = 1/4$ — 1

$\therefore$ Forced response is $\dfrac{1}{4(s+3)}$

c. (5 points) Calculate the initial value $y(0+)$ of this system's response.

1 point for trying

Using the initial value theorem

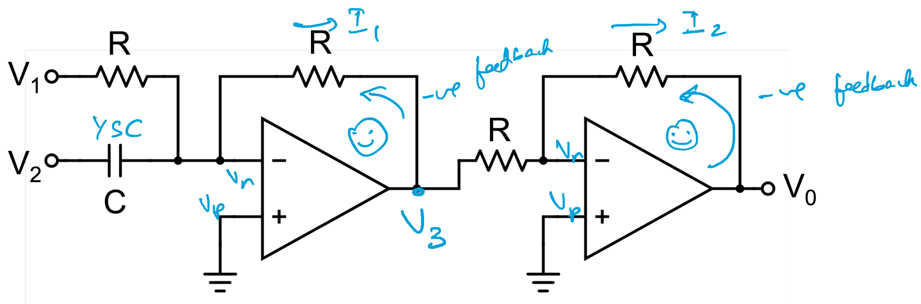
$$y(0^+) = \lim_{s\to\infty} sY(s) \quad — 1$$

$$\lim_{s\to\infty} sY(s) = \underbrace{\lim_{s\to\infty} \frac{s(s+4)}{(s+1)^2(s+3)}}_{1} = \lim_{s\to\infty} \frac{\frac{1}{s}\left(1+\frac{4}{s}\right)}{\left(1+\frac{1}{s}\right)^2\left(1+\frac{3}{s}\right)}$$

$$= 0 \quad — 2$$

5. (10 points) Operational Amplifiers

For the circuit below, there was no energy stored initially in the capacitor.



a. (6 points) Find the output voltage $V_0(s)$ in terms of input voltages $V_1(s)$ and $V_2(s)$ and the circuit elements R and C.

Both opamps operate in -ve feedback mode

$\Rightarrow$ stable $\Rightarrow$ $V_p = V_n$ — (1)

1 point for trying

$V_p = 0$ [ground] $\therefore$ $V_n = 0$ (1)

$$I_1 = \frac{V_2 - 0}{1/sC} + \frac{V_1 - 0}{R} = V_2 sC + \frac{V_1}{R} \quad (1)$$

$$V_3 = 0 - I_1 R = -R\left(\frac{V_1}{R} + V_2 sC\right) = -(V_1 + V_2 sRC) \quad (1)$$

$$V_0 = -\frac{R}{R} V_3 = -V_3 = (V_1 + V_2 sRC) \quad (1)$$

b. (4 points) If $R = 1K\Omega$, $C = 0.1mF$, find the value of s for which $V_0(s) = V_1(s) + V_2(s)$?

$$V_0(s) = V_1(s) + V_2(s) \cdot sRC$$

$$= V_1(s) + V_2(s) \quad \text{if} \quad sRC = 1 \quad — (1)$$

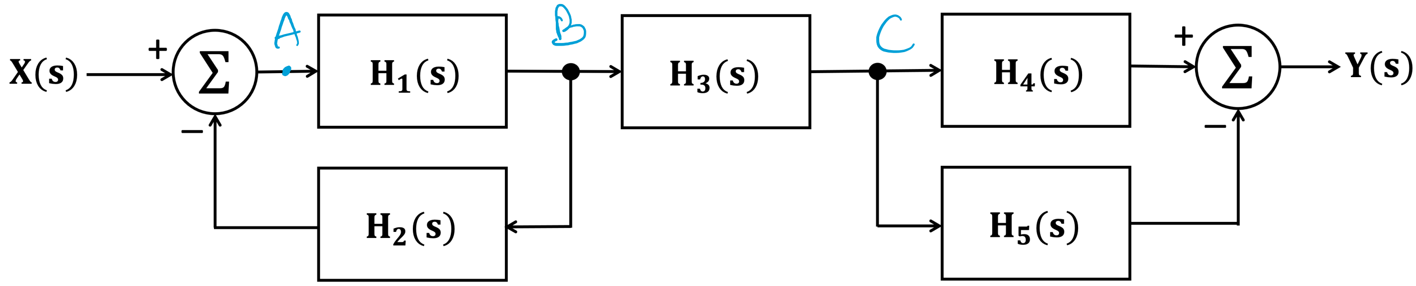
1 point for trying

$$s = \frac{1}{RC} = \frac{1}{1K\Omega \times 0.1\,mf} = \frac{1}{0.1} = 10\ Hz \quad (1)$$

(1)

6. (10 points) System Block Diagrams

Calculate the overall transfer function $\mathbf{H}(\mathbf{s})$ for the following connection of LTI systems. Your final answer should be expressed in the Laplace domain.



$A = X - B H_2$ — 1

$B = A \cdot H_1$ — 1

$C = B H_3$ — 1

$Y = C H_4 - C H_5$ — 1

2 points for trying

$A = X - A H_1 H_2$ — 1

$A(1 + H_1 H_2) = X$

$A = \dfrac{X}{1 + H_1 H_2}$ — 1

$C = B H_3 = A H_1 H_3 = \dfrac{H_1 H_3}{1 + H_1 H_2}$ — 1

$Y = (H_4 - H_5) \dfrac{H_1 H_3}{1 + H_1 H_2}$ — 1

Full points for correct answer